

Factoring Algebraic Expressions (Part IV) Homework

Factor completely.

① $3x^4 + 9x^3 - 30x^2$

② $5y^3 - 20y$

③ $-4b^5 + 28b^4 - 24b^3$

④ $5x^2 - 45$

⑤ $-7y^3 + 7y$

⑥ $b^3 + 5b^2 + 6b$

⑦ $10x^{10} - 10x^9 - 120x^8$

⑧ $6y^3 - 36y^2 + 54y$

⑨ $8b^7 - 128b^5$

⑩ $2x^4 - 4x^3 + 4x^2 - 8x$

⑪ $5y^7 + 15y^6 - 15y^5 - 45y^4$

⑫ $-2b^6 - 6b^4 - 4b^2$

⑬ $4x^5 - 4x^3 - 48x$

⑭ $y^5 - 16y$

⑮ $-2b^7 + 12b^5 - 10b^3$

⑯ $2x^{11} + 3x^{10} - 9x^9$

⑰ $90x^5 - 1000x^3$

⑱ $-6x^9 + x^8 + 12x^7$

⑲ $5x^4 + 20x^3y + 20x^2y^2$

⑳ $2a^3b^2 - 6a^2b^3 - 36ab^4$

㉑ $3x^2y - 243y^3$

㉒ $4a^3b + 23a^2b^2 - 6ab^3$

㉓ $-8x^2y^2 - 16xy^3 + 120y^4$

㉔ $11a^5b^2 - 44a^3b^4$

Answers

- ① $3x^2(x+5)(x-2)$
- ② $5y(y-2)(y+2)$
- ③ $-4b^3(b-6)(b-1)$
- ④ $5(x+3)(x-3)$
- ⑤ $-7y(y+1)(y-1)$
- ⑥ $b(b+2)(b+3)$
- ⑦ $10x^8(x-4)(x+3)$
- ⑧ $6y(y-3)(y-3)$
- ⑨ $8b^5(b-4)(b+4)$
- ⑩ $2x(x^2+2)(x-2)$
- ⑪ $5y^4(y^2-3)(y+3)$
- ⑫ $-2b^2(b^2+1)(b^2+2)$
- ⑬ $4x(x^2+3)(x+2)(x-2)$
- ⑭ $y(y+2)(y-2)(y^2+4)$
- ⑮ $-2b^3(b-1)(b+1)(b^2-5)$
- ⑯ $x^9(2x-3)(x+3)$
- ⑰ $10x^3(3x-10)(3x+10)$

- ⑱ $-x^7(3x+4)(2x-3)$
- ⑲ $5x^2(x+2y)(x+2y)$
- ⑳ $2ab^2(a+3b)(a-6b)$
- ㉑ $3y(x-9y)(x+9y)$
- ㉒ $ab(4a-b)(a+6b)$
- ㉓ $-8y^2(x-3y)(x+5y)$
- ㉔ $11a^3b^2(a-2b)(a+2b)$